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Product Sales Analysis

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

Code:

```
# Load
```

```
libraries
```

```
library(ggplot2
```

```
) library(tidyr)
```

```
# Create
```

```
dataset sales <-
```

```
data.frame(
```

```
  Product_ID = c(1,2,3),
```

```
  Product_Name = c("Product A","Product B","Product C"),
```

```
  January = c(2000,1500,1200),
```

```
  February = c(2200,1800,1400),
```

```
  March = c(2400,1600,1100)
```

```
)
```

```
# Convert to long format
```

```
sales_long <- pivot_longer(
```

```
sales, cols = c(January,
```

```
February, March),
```

```
names_to = "Month",
```

```
values_to = "Sales"
```

```
)
```

```
# Grouped Bar Chart

ggplot(sales_long,
aes(x =
Product_Name,
      y = Sales,      fill = Month))
+ geom_bar(stat = "identity", position
= "dodge") + labs(  title = "First
Quarter Product Sales",  x =
"Product",  y = "Sales"
) +
theme_minimal()
```

Output:



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

Program:

```
# Load
libraries
```

```

library(ggplot2
) library(tidyr)
# Create
dataset sales <-
data.frame(
  Product_Name = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)
# Convert to long format
sales_long <- pivot_longer(
sales, cols = c(January,
February, March),
names_to = "Month",
values_to = "Sales"
)
# Order months
sales_long$Month <-
factor( sales_long$Month,
levels =
c("January","February","M
arch")
)
# Stacked Area Chart
ggplot(sales_long,

```

```

aes(x = Month,      y =
Sales,      fill =
Product_Name,
group = Product_Name))
+ geom_area(alpha =
0.8) +
labs(  title = "Overall
Sales Trend",  x =
"Month",  y = "Sales"
) +
theme_minimal()

```

Output:



3. Build a table to show the monthly sales data for each product. Label the table elements.

Program:

```

# Create dataset

sales <-
data.frame(
  Product_ID = c(1,2,3),

```

```

Product_Name = c("Product A","Product B","Product C"),
January_Sales = c(2000,1500,1200),
February_Sales = c(2200,1800,1400),
March_Sales = c(2400,1600,1100)
)
# Display table
print(sales)

```

Output:

```

Product_Name January February March
1 Product A    2000    2200  2400
2 Product B    1500    1800  1600
3 Product C    1200    1400  1100

```

Set 7: Customer Demographics Analysis

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

Program:

```

# Load library
library(ggplot2) #
Create dataset
customers <-
data.frame(
  Customer_ID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

```

```
# Bar Chart ggplot(customers,
aes(x = factor(Customer_ID),
y = Age, fill =
factor(Customer_ID))) +
geom_bar(stat = "identity") +
labs(title = "Customer Age
Distribution", x =
"Customer ID", y = "Age"
) +
theme_minimal()
```

Output:



2. Generate a pie chart to represent the distribution of customers by gender.

Program:

```
# Load libraries
library(ggplot2) #
Create dataset
customers <-
data.frame(
Customer_ID = c(1,2,3),
Gender = c("Female","Male","Female")
)
```

```

)
# Count genders gender_count <-
as.data.frame(table(customers$Gender))
colnames(gender_count) <-
c("Gender","Count")
# Pie Chart
ggplot(gender_count,
       aes(x = "", y = Count,
           fill = Gender)) + geom_bar(stat =
"identity", width = 1) +
coord_polar("y") + labs(title =
"Customer Gender Distribution") +
theme_void()

```

Output:



3. Build a table to show the demographic information for each customer. Label the table elements.

Program:

```
# Create dataset

customers <-
data.frame(
  Customer_ID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

# Display table
print(customers)
```

Output:

```
Customer_ID Gender
1          1 Female
2          2  Male
3          3 Female
```

Set 8: Employee Performance Analysis

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

Program:

```
# Load library
library(ggplot2)

# Create dataset
employee <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
```

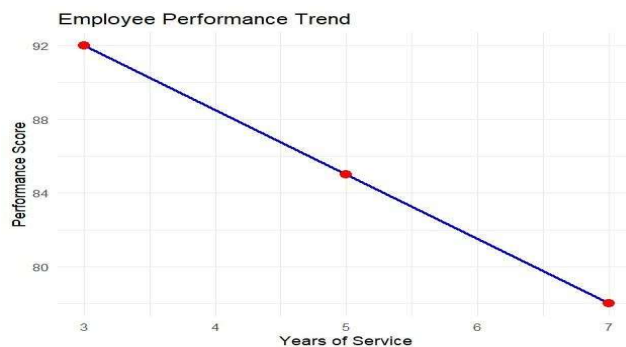


```

Performance_Score = c(85, 92, 78)
)
# Line Chart ggplot(employee, aes(x = Years_of_Service, y =
Performance_Score, group = 1)) + geom_line(color = "blue",
linewidth = 1) + geom_point(color = "red", size = 3) + labs(
title = "Employee Performance Trend", x = "Years of Service",
y = "Performance Score"
) +
theme_minimal()

```

Output:



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

Program:

```

# Load library
library(ggplot2) #
Create dataset
employee <-
data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),

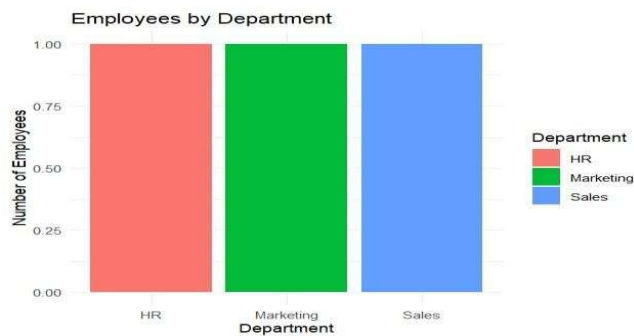
```

```

Performance_Score = c(85, 92, 78)
)
# Bar Chart
ggplot(employee,
aes(x = Department,
fill = Department)) +
geom_bar() +
labs( title = "Employees
by Department", x =
"Department", y =
"Number of Employees"
) +
theme_minimal()

```

Output:



3. Build a table to display the performance data for each employee. Label the table elements.

Program:

```

# Create dataset
employee <-
data.frame(
Employee_ID = c(1, 2, 3),

```

```

Department = c("Sales", "HR", "Marketing"),
Years_of_Service = c(5, 3, 7),
Performance_Score = c(85, 92, 78)
)
# Display table
print(employee)

```

Output:

	Employee_ID	Department	Years_of_Service	Performance_Score
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

set 9: Online_Learning_Activity.csv

1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

Program:

```

# Load library
library(ggplot2) #
Create dataset
learning <-
data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Course = c("R","R","SQL","R","R","SQL"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),

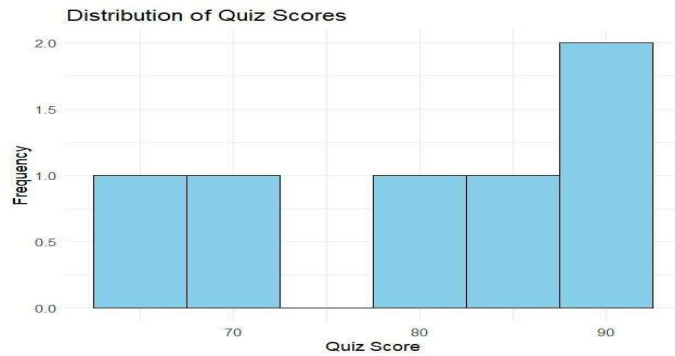
```

```

Videos_Watched = c(12,15,8,18,9,14),
Quiz_Score = c(78,85,65,92,70,88),
Login_Date = c("2025-01-05","2025-01-05","2025-02-08",
               "2025-02-08","2025-03-12","2025-03-12")
)
# Histogram ggplot(learning, aes(x = Quiz_Score)) +
geom_histogram(binwidth = 5, fill = "skyblue",
color = "black") + labs(title = "Distribution of
Quiz Scores", x = "Quiz Score", y =
"Frequency"
) +
theme_minimal()
# Boxplot
ggplot(learning,
aes(x = Course,
y = Quiz_Score,
fill = Course)) +
geom_boxplot() +
labs(title = "Quiz
Scores by Course", x =
"Course", y = "Quiz
Score"
) +
theme_minimal()

```

Output:



2. Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent

Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

Program:

```
# Load library
```

```
library(ggplot2) #
```

```
Create dataset
```

```
learning <-
```

```
data.frame(
```

```
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
```

```
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),
```

```
  Quiz_Score = c(78, 85, 65, 92, 70, 88),
```

```
  Videos_Watched = c(12, 15, 8, 18, 9, 14)
```

```
)
```

```
# Bubble Scatter Plot
```

```
ggplot(learning, aes(x =
```

```
  Study_Time, y =
```

```
  Quiz_Score, size =
```

```
  Videos_Watched)) +
```

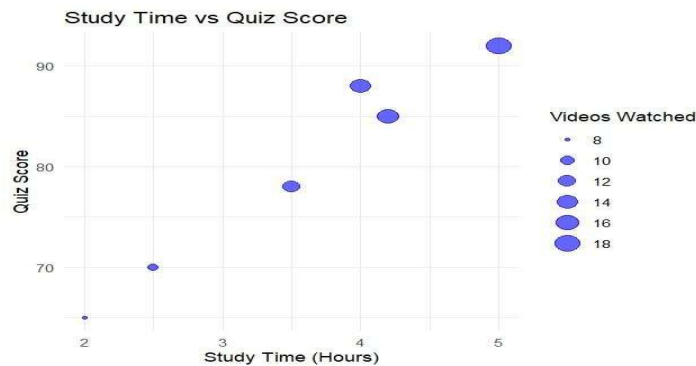
```
geom_point(alpha = 0.6, color =
```

```

"blue") + labs( title = "Study
Time vs Quiz Score", x =
"Study Time (Hours)", y =
"Quiz Score", size = "Videos
Watched"
) +
theme_minimal()

```

Output:



3. Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

Program:

```

# Load libraries

library(ggplot2)

library(dplyr) #

Create dataset

learning <-
data.frame(

  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Quiz_Score = c(78,85,65,92,70,88),

```

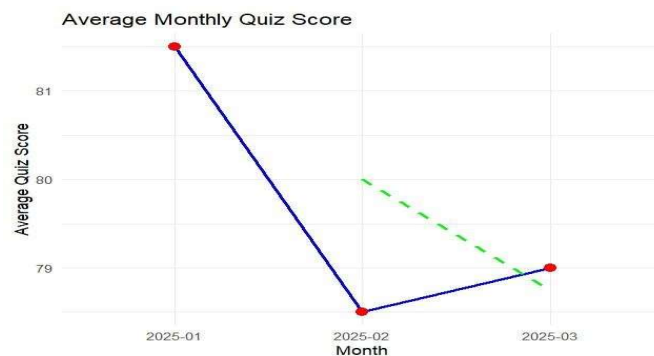
```

Login_Date = c("2025-01-05","2025-01-05","2025-02-08",
               "2025-02-08","2025-03-12","2025-03-12")
)
# Convert to Date learning$Login_Date <-
as.Date(learning$Login_Date)
# Extract Month learning$Month <-
format(learning$Login_Date, "%Y-%m")
# Average Quiz Score per Month
monthly_avg <- learning %>%
group_by(Month) %>%
summarise(Average_Score =
mean(Quiz_Score))
# Moving Average
monthly_avg$Moving_Average <-
stats::filter(
monthly_avg$Average_Score,
rep(1/2, 2),
sides = 1
)
# Line Chart ggplot(monthly_avg, aes(x = Month, y =
Average_Score, group = 1)) + geom_line(color =
"blue", linewidth = 1) + geom_point(size = 3, color =
"red") + geom_line(aes(y = Moving_Average),
                    color = "green",
linewidth = 1,      linetype =
"dashed") + labs(  title =

```

```
"Average Monthly Quiz Score",
x = "Month", y = "Average
Quiz Score"
) +
theme_minimal()
```

Output:



Set 10: Survey Responses Analysis

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

Program:

```
# Load library
library(ggplot2)

# Create dataset
survey <-
data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)
```



```

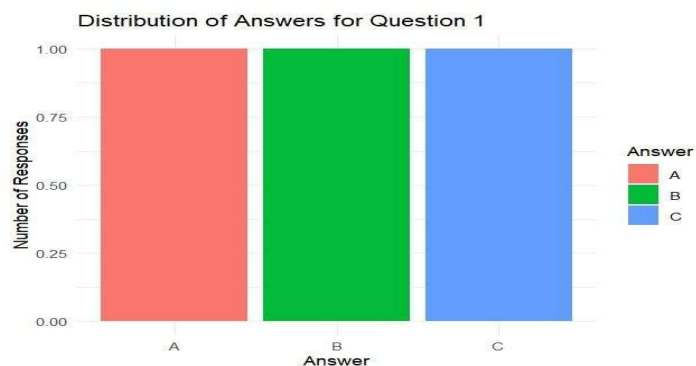
# Count responses for Question 1

q1 <-
as.data.frame(table(survey$Question1))
colnames(q1) <- c("Answer",
"Count")

# Grouped Bar Chart
ggplot(q1, aes(x =
Answer, y = Count, fill = Answer)) +
geom_bar(stat = "identity") + labs( title
= "Distribution of Answers for Question 1",
x = "Answer", y = "Number of
Responses"
) +
theme_minimal()

```

Output:



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

Program:

```

# Load libraries

library(ggplot2)

library(tidyr) #
Create dataset

```

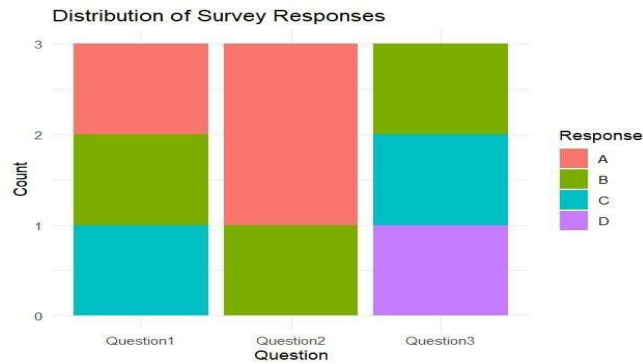
```

survey <-
data.frame(
  Survey_ID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
# Convert to long format
survey_long <- pivot_longer(
  survey, cols = c(Question1,
  Question2, Question3), names_to
= "Question", values_to =
"Response"
)
# Stacked Bar Chart
ggplot(survey_long,
aes(x = Question,
fill = Response)) +
geom_bar() +

  labs( title = "Distribution of
Survey Responses", x =
"Question", y = "Count"
) +
  theme_minimal()

```

Output:



3. Build a table to show the survey response data for each survey. Label the table elements.

Program:

```
# Create dataset
```

```
survey <-
```

```
data.frame(
```

```
  Survey_ID = c(1,2,3),
```

```
  Question1 = c("A","B","C"),
```

```
  Question2 = c("B","A","A"),
```

```
  Question3 = c("C","D","B")
```

```
)
```

```
# Display table
```

```
print(survey)
```

Output:

```
Survey_ID Question1 Question2 Question3
```

```
1      1      A      B      C
```

```
2      2      B      A      D
```

```
3      3      C      A      B
```

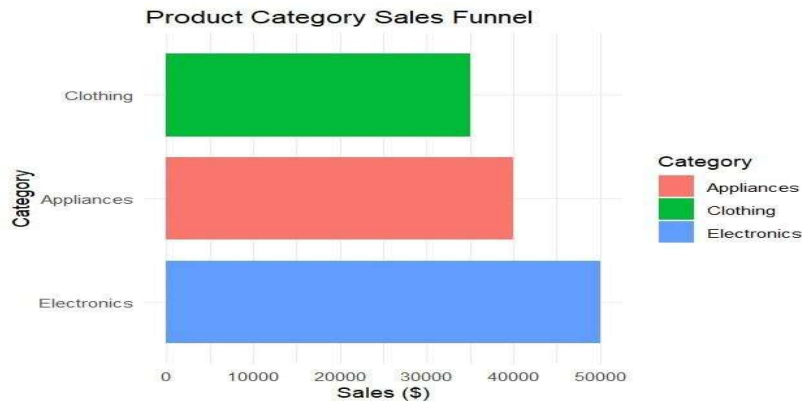
Set 11: Product Category Analysis

1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

Program:

```
# Load library
library(ggplot2) #
Create dataset
category <-
data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(500000, 350000, 400000)
)
# Funnel Chart ggplot(category,
aes(x = reorder(Category, -Sales),
y = Sales,      fill = Category))
+ geom_bar(stat = "identity",
width = 0.8) + coord_flip() +
labs(  title = "Product Category
Sales Funnel",  x = "Category",
y = "Sales ($)"
) +
theme_minimal()
```

Output:



2. Build a table to display the sales data for each product category. Label the table elements.

Program:

```
# Create dataset

category <-
data.frame(
  Category = c("Electronics","Clothing","Appliances"),
  Sales = c(50000,35000,40000)
)

# Display table

print(category)
```

Output:

Category Sales

- 1 Electronics 50000
- 2 Clothing 35000
- 3 Appliances 40000

3. Create a pie chart to represent the distribution of sales across product categories. Include labels.

Program:

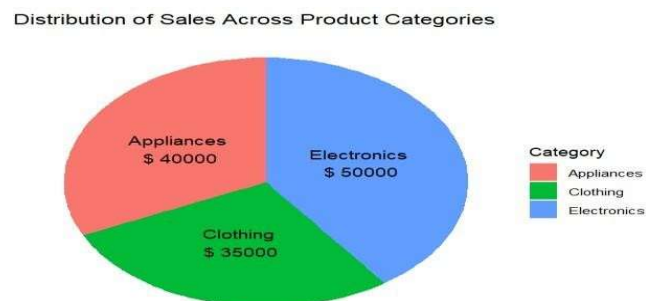
```
# Load library
```

```
library(ggplot2) #
```

Create dataset

```
category <-  
data.frame(  
  Category = c("Electronics", "Clothing", "Appliances"),  
  Sales = c(50000, 35000, 40000)  
)  
# Pie Chart ggplot(category, aes(x = "", y =  
Sales, fill = Category)) + geom_bar(stat =  
"identity", width = 1) + coord_polar(theta  
= "y") + geom_text(  aes(label =  
paste(Category, "\n$", Sales)),  position =  
position_stack(vjust = 0.5),  size = 4  
) +  
labs(  title = "Distribution of Sales Across  
Product Categories"  
) +  
theme_void()
```

Output:



Set 12: Dataset: Website Traffic

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

Program:

```
# Load library

library(ggplot2)

# Create dataset

traffic <-

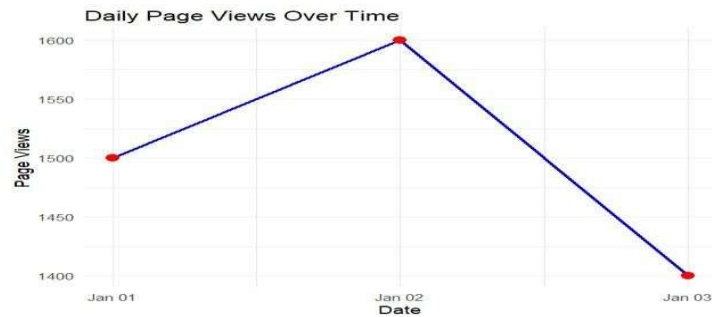
data.frame(

  Date = as.Date(c("2023-01-01",
                   "2023-01-02",
                   "2023-01-03")),
  Page_Views = c(1500,1600,1400),
  Click_Through_Rate = c(2.3,2.7,2.0)
)

# Line Chart ggplot(traffic,
aes(x = Date,      y =
Page_Views,      group = 1)) +
geom_line(color = "blue",
linewidth = 1) +
geom_point(color = "red", size =
3) +

labs(  title = "Daily Page
Views Over Time",  x =
"Date",  y = "Page Views"
) +
theme_minimal()
```

Output:



2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

Program:

```
# Load library
library(ggplot2)

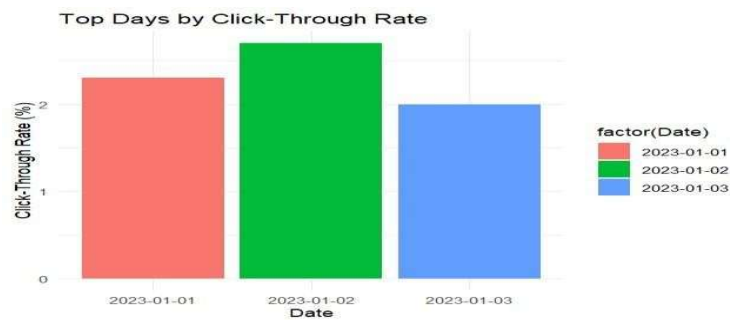
# Create dataset
traffic <-
data.frame(
  Date = as.Date(c("2023-01-01",
                   "2023-01-02",
                   "2023-01-03")),
  Click_Through_Rate = c(2.3,2.7,2.0)
)

# Sort data top_days <- traffic[order(-
traffic$Click_Through_Rate), ]

# Bar Chart
ggplot(top_days,
aes(x = factor(Date),
y =
Click_Through_Rate,
fill = factor(Date))) +
```

```
geom_bar(stat =
"identity") + labs(
title = "Top Days by
Click-Through Rate",
x = "Date", y =
"Click-Through Rate
(%)"
) +
theme_minimal()
```

Output:



3. Build a table to show the website traffic data for each day. Label the table elements.

Program:

```
# Create dataset

traffic <-
data.frame(
  Date = c("2023-01-01",
            "2023-01-02",
            "2023-01-03"),
  Page_Views = c(1500,1600,1400),
  Click_Through_Rate = c("2.3%","2.7%","2.0%")
)
```

```
# Display table
```

```
print(traffic)
```

Output:

	Date	Click_Through_Rate
1	2023-01-01	2.3
2	2023-01-02	2.7
3	2023-01-03	2.0